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Short communication

# Artificial regulation of gradient heterostructures in CoCrNi multicomponent alloys via electron beam surface melting

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ABSTRACT

Gradient heterostructure design effectively balances strength and ductility, yet its precise control remains challenging. By applying electron beam surface melting (EBSM) to a cold-rolled CoCrNi multicomponent alloy, three distinct microstructural zones, i.e., coarse columnar grains, fine equiaxed grains, and deformed structures, are artificially regulated. The hardness varies continuously from the surface to the center, and the EBSM-treated sample achieves an optimal balance between strength and ductility under controlled conditions. The underlying strengthening mechanisms can be attributed to the rule of mixtures and hetero-deformation-induced (HDI) strengthening. This work provides valuable insights into the artificial regulation of gradient heterostructures for high-performance alloys.

## 1. Introduction

Achieving a harmonious strength-ductility balance in metals and alloys has been a critical challenge in materials science [1–7]. To optimize overall performance, researchers have explored various strategies [1–7], with heterostructured materials emerging as a promising solution by introducing structural inhomogeneities at the microscale or nanoscale [8–15]. Different design strategies, such as bimodal structures, lamellar structures, gradient structures, multiphase structures, harmonic structures, and metal matrix composites, have been proposed [1–15]. The coexistence of hard and soft zones [10,16–18] contributes to their unique mechanical properties. Among them, gradient heterostructures are particularly attractive due to their good strength-ductility balance [17,19–23]. The back stress generated between hard and soft zones provides additional hetero-deformation-induced (HDI) hardening that differs from conventional mechanisms [24,25]. Therefore, designing gradient heterostructures represents a promising strategy to address the strength-ductility trade-off [11,18,25].

Currently, methods to achieve gradient heterostructures include ultrasonic surface rolling [26], rotationally accelerated shot peening [21,27], and surface mechanical abrasion treatments [28]. However, these techniques often involve complex experimental procedures and extended processing times [25,26,29–32]. In contrast, thermal modulation methods such as electromagnetic induction heating or laser surface treatment offer more efficient approaches [33–35]. For example, the laser scanning of cold-rolled sheets produces gradient heterostructures [36], though the relatively small heat-affected zone (HAZ) limits performance improvements [37]. In this work, we introduce an innovative approach to engineer gradient heterostructures using representative equiatomic CoCrNi multicomponent alloys (MAs) as the base material. Our methodology combines cold rolling with fast electron-beam surface melting (EBSM) to induce microstructural transitions through localized melting and heating effects, yielding advanced material configurations with tailored properties.

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## 2. Experimental

The equiatomic CoCrNi ingots (>99.9 %) were fabricated via arc melting and the corresponding plates were prepared through suction casting. The as-cast plates (50 × 10 × 9.3 $mm^3$) were homogenized at 1423 K for 24 h and water-quenched. Afterward, the plates were subsequently cold-rolled at a speed of 100 r/min, achieving an 80 % thickness reduction through successive passes of 0.25 mm each. Fast electron-beam surface melting (EBSM) was then performed on the cold-rolled samples under a vacuum environment at an accelerating voltage of 85 kV with a focusing current of 420 mA. For the E1, E2, E3, and E4 samples, the electron beam (E-beam) currents were set to 1 mA, 2 mA, 3 mA, and 4 mA, respectively. Both the upper and lower surfaces of each sample were scanned unidirectionally at a constant scanning speed of 20 mm/s, with a scanning interval of 0.5 mm and a 30-s pause between scans. After 25 scans, the total beam energy densities were calculated to be 773 ± 50, 1006 ± 50,1079 ± 50, 1088 ± 50 W/$cm^2$, respectively. Residual internal stress measurement by X-ray diffraction (XRD, proto iXRD). to characterize the microstructure, laser scanning confocal microscopy (LSCM, Keyence VK-X1000) was employed to measure the surface roughness and morphologies of the cross-section (right inset in Fig. 1(a)). Electron backscatter diffraction (EBSD, Oxford Nordlys max$^3$) with a step size of 0.55 μm was also used. The EBSD samples were mechanically ground, polished with Reflex PAD-MAG and diamond suspensions (3 μm/1 μm), vibratory polished with SUPRA cloth and $SiO_2$ (0.03 μm) at 0.63 mm, 42 Hz for 16 h, and etched for 5–10 s with 7.5 g $CuSO_4{\cdot}5H_2O$, 20 mL HCl, and 2 mL $H_2SO_4$. Tensile testing was conducted on flat dog-bone-shaped specimens with a gauge dimension of 2.5 × 1.85 × 6 $mm^3$ at a strain rate of 1 × $10^{-3}$ $s^{-1}$. Vickers hardness measurements were performed using an MHV-1000Z tester under a load of 1.961 N applied for 10 s.

## 3. Results and discussion

### 3.1. Microstructural features

Fig. 1(b–e) shows the structural features of the EBSM-treated E1, E2, E3, and E4 samples. All EBSM-treated samples exhibit parallel melting

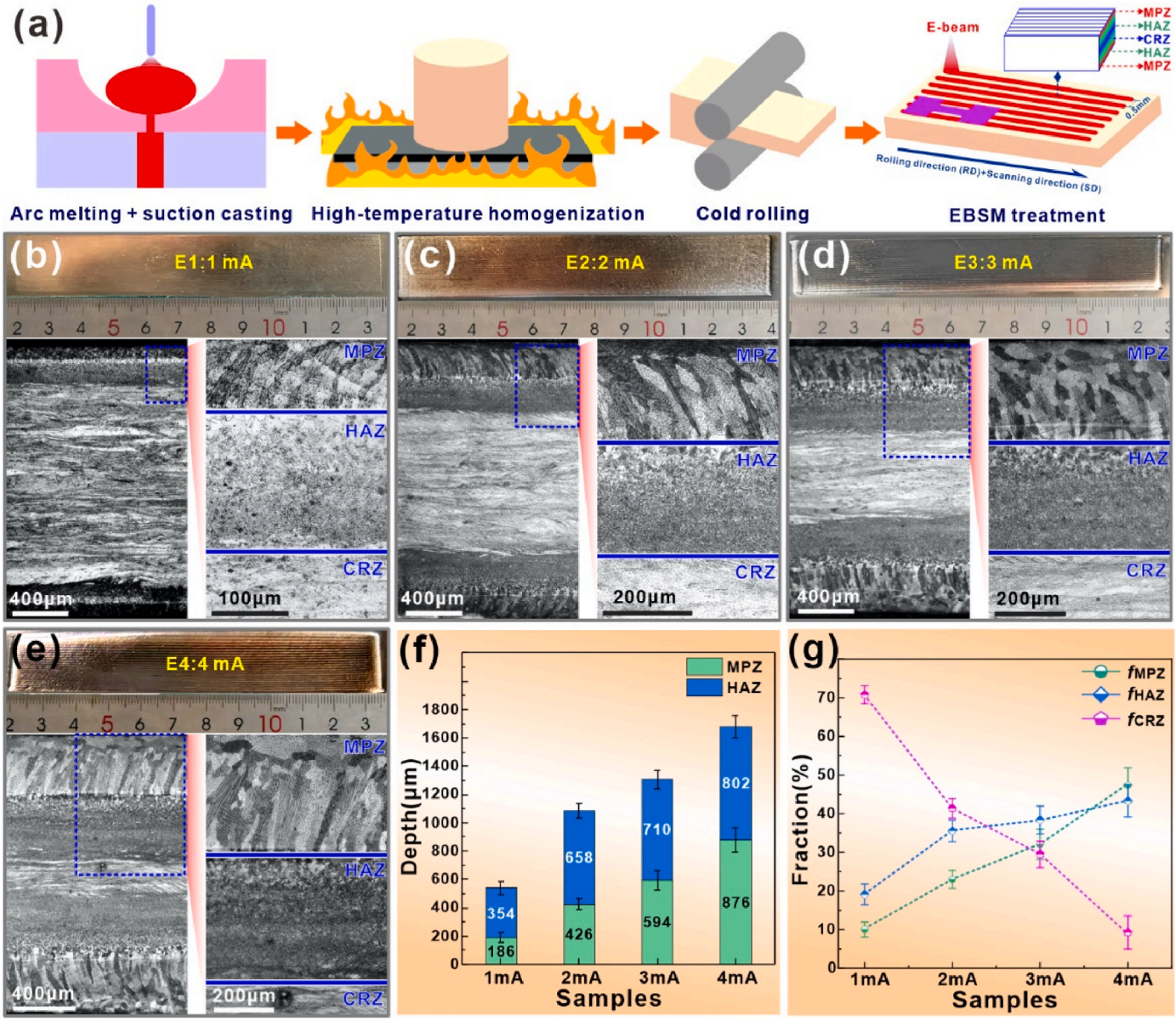



**Fig. 1.** (a) Sample preparation process diagram; LSCM images of the (b) E1, (c) E2, (d) E3, and (e) E4 samples; (f) the depths of MPZ and HAZ of different samples and (g) the volume fractions of different zones.

traces on their surfaces, which become more prominent with increasing the E-beam current. The surface roughness of these samples above is measured to be 16 ± 4 μm, 17 ± 5 μm, 14 ± 4 μm, and 17 ± 7 μm, respectively. Three distinct microstructural zones from the surface to the center can be observed, namely the melt pool zone (MPZ), the HAZ [37,38], and the original cold-rolled zone (CRZ), resulting in noticeable gradient heterostructural morphologies and symmetrical features. The proportions of these zones are influenced by variations in the E-beam current. As shown in Fig. 1(b–f), increasing the E-beam current enlarges the MPZ depth from 93 ± 8 μm to 438 ± 20 μm and the HAZ depth from 177 ± 12 μm to 401 ± 20 μm, leading to a decrease in the volume fraction of the CRZ. Fig. 1(g) demonstrates significant changes in the proportions of these zones. The CRZ decreases from about 70.81 % to 9.30 %, the MPZ increases from about 10.05 % to 47.35 %, and the HAZ rises from about 19.14 % to 43.35 %, resulting in the almost disappearance of the CRZ. The heat distribution in the MPZ promotes grain growth in a slightly vertical direction, while the dissipated heat facilitates the recrystallization of the HAZ [37]. It was reported that cold rolling forms an $\alpha$-fiber structure with stacking faults, deformation twins, and shear bands in the CoCrNi MAs [39,40], and the CRZ remains largely unaffected, retaining its textured structure. Thus, by adjusting the E-beam current, the proportions of these zones can be precisely controlled, enabling the manipulation of gradient heterostructures.

To further understand the microstructure evolution from the surface to the center, EBSD investigations were conducted (Fig. 2(a)). The MPZ consists of equiaxed grains and prominent coarse columnar grains in the outermost layer [41–44]. As the E-beam current increases, the equiaxed grains in the outermost layer become more evident and coarser, likely due to a higher solidification rate and smaller temperature gradient [44]. Inside the MPZ, columnar grains grow from the melt pool

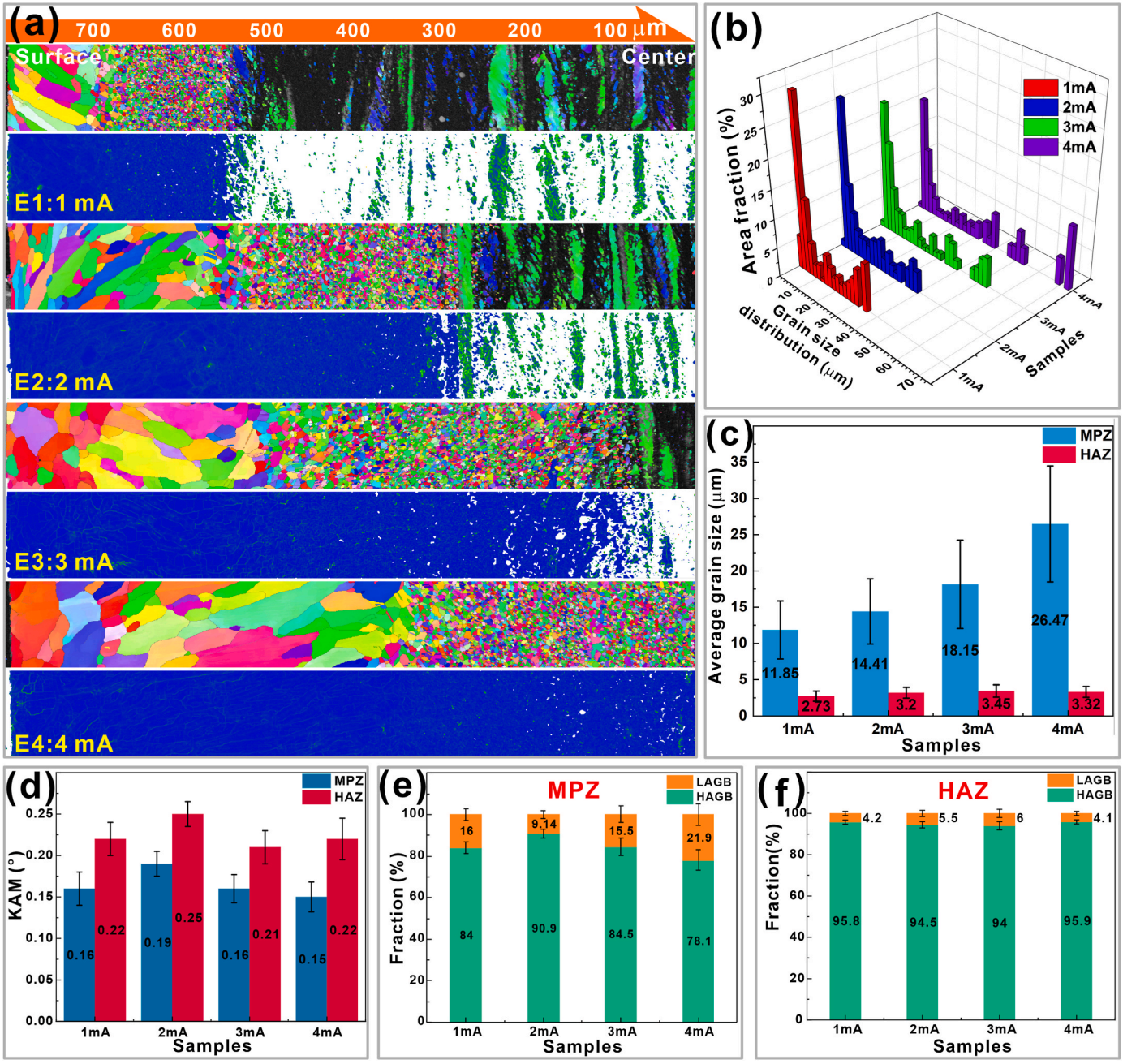



**Fig. 2.** (a) IPF and KAM maps from the surface to the center of samples E1, E2, E3, and E4; (b) grain size distribution, (c) average grain size, and (d) average KAM values in the MPZ and HAZ; the proportions of LAGBs and HAGBs in the (e) MPZ and (f) HAZ.

boundary toward the center, aligning with the direction of maximum heat flow. There is no twin formation within these grains because fast solidification restricts elemental diffusion [43,45]. The HAZ also exhibits a gradient structure, transitioning from micron-sized coarse grains to submicron grains, and contains annealing twins (Fig. 2(a)). The inverse pole figure (IPF) in Fig. 2(a) indicates that the HAZ is predominately characterized by fine equiaxed grains, confirming the recrystallization of the CRZ. The recrystallization degree in the HAZ increases from the interface between the CRZ and MPZ, and this trend becomes more pronounced with a higher E-beam current. Fig. 2(b) illustrates the overall grain size distribution in the MPZ and HAZ under different E-beam currents. This distribution spans several orders of magnitude. As the E-beam current increases, the proportion of coarse grains (≥10 μm) rises, while the proportion of fine grains (<10 μm) decreases. The maximum grain size also increases from approximately 35 μm–70 μm. Fig. 2(c) shows that the average grain size in the MPZ increases from approximately 11.85 μm–26.47 μm. In the HAZ, the average grain size grows from about 2.73 μm to 3.45 μm and then stabilizes. Fig. 2(a) and (d) reveal that Kernel average misorientation (KAM) values in the MPZ and the HAZ are lower compared to those in the CRZ. The HAZ has a relatively higher KAM value than the MPZ. The KAM value typically indicates the GND density, suggesting that both regions have low dislocation density around grain boundaries, but the HAZ has a higher dislocation density than the MPZ. Additionally, the grain boundary distribution (Fig. 2(e) and (f)) shows that the MPZ and HAZ are primarily characterized by high-angle grain boundaries (HAGBs). In the MPZ, as the E-beam current increases, the fraction of low-angle grain boundaries (LAGBs) initially decreases from ~16.0 % to ~9.14 % and then rises to ~21.9 %. In contrast, the HAZ always maintains a high HAGB proportion of more than 94 %, indicating a higher recrystallization.

### 3.2. Mechanical performance and multiple strengthening mechanisms

Fig. 3(a) shows the mechanical properties of the gradient heterostructured samples. The cold-rolled sample exhibits an average microhardness of 462 ± 15 HV, attributed to the high density of dislocations, grain boundaries, and textures. After EBSM treatments, the microhardness decreases from the center to the surface, reflecting the gradient structure, which becomes more pronounced with increasing E-beam current. In the MPZ, the measured microhardness is 225 ± 25 HV, primarily due to the relatively coarse grain size. In the HAZ, the microhardness ranges from 225 ± 25 HV to 462 ± 15 HV, indicating varying recrystallization degrees induced by EBSM treatments. Fig. 3(b) and (c) show the tensile curves of the homogenized, cold-rolled, and EBSM-treated samples. As the E-beam current increases, the yield strength decreases from approximately 1284 MPa–675 MPa, while the total elongation increases from about 11.3 % to 37 %. Compared with the homogenized sample, the gradient heterostructured samples exhibit higher yield strength and ultimate tensile strength, along with acceptable ductility. Fig. 3(d) indicates that the strain-hardening rate of the cold-rolled sample drops rapidly to zero, suggesting strong strain-softening behavior. After EBSM treatments, the strain-hardening rate increases with the E-beam current, transitioning from strain-softening to strain-hardening behavior. These observations suggest that EBSM treatments can effectively modify the gradient heterostructure and enhance the mechanical properties.

The yield strength ($\sigma_y$) of a metallic material is influenced by several factors, including lattice friction stress ($\sigma_i$), solid solution strengthening ($\Delta\sigma_{ss}$), grain boundary strengthening ($\Delta\sigma_{gb}$), precipitation strengthening ($\Delta\sigma_p$), strengthening contribution from the pre-existing dislocations ($\Delta\sigma_d$), and texture strengthening ($\Delta\sigma_t$). In our case, no precipitates are observed in the equiatomic CoCrNi MAs, so the $\Delta\sigma_p$ is considered zero. The combined effect of the $\sigma_i$ and $\Delta\sigma_{ss}$ is approximately 218 MPa, as $\Delta\sigma_{ss}$ is incorporated into $\sigma_i$ for the CoCrNi MAs [46,47]. Given the three distinct zones, the current MAs can be treated as a composite to better understand their strengthening mechanisms. The $\sigma_y$ is calculated using the rule of mixtures (ROM):

$$\sigma_y = \sigma_{MPZ} \times f_{MPZ} + \sigma_{HAZ} \times f_{HAZ} + \sigma_{CRZ} \times f_{CRZ}, \quad (1)$$

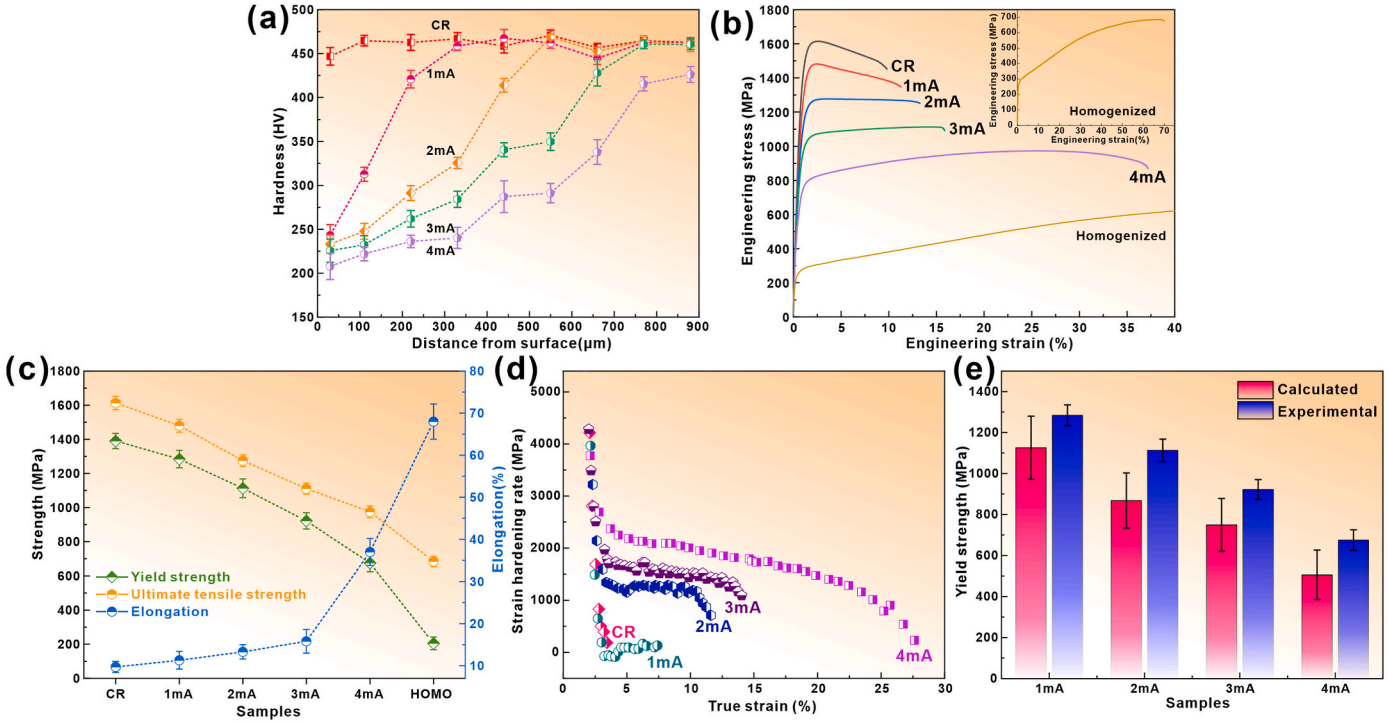



**Fig. 3.** (a) Changes in hardness, (b) engineering stress-strain curves of homogenized, cold-rolled, and EBSM-treated samples (Inset: the overall stress-strain curve of the homogenized sample), (c) changes in yield strength, ultimate tensile strength, and total elongation, and (d) the strain-hardening rates, and (e) the calculated and experimental $\sigma_y$.

where $\Delta\sigma_{MPZ}$, $\Delta\sigma_{HAZ}$, and $\Delta\sigma_{CRZ}$ represent the strengthening contributions of the MPZ, HAZ, and CRZ, respectively, and $f_{MPZ}$, $f_{HAZ}$, and $f_{CRZ}$ are their volume fractions (Fig. 1(g)), which sum to 1. The strengthening contributions of the MPZ and HAZ ($\Delta\sigma_{MPZ}$ and $\Delta\sigma_{HAZ}$) are evaluated by

$$\sigma_{MPZ} = \sigma_i + \Delta\sigma_{ss} + \Delta\sigma_{gb,MPZ} + \Delta\sigma_{d,MPZ}, \quad (2)$$

$$\sigma_{HAZ} = \sigma_i + \Delta\sigma_{ss} + \Delta\sigma_{gb,HAZ} + \Delta\sigma_{d,HAZ}, \quad (3)$$

The strengthening contribution of the CRZ ($\Delta\sigma_{CRZ}$) can be estimated by

$$\sigma_{CRZ} = \sigma_i + \Delta\sigma_{ss} + \Delta\sigma_{gb,CRZ} + \Delta\sigma_{CRZ} + \Delta\sigma_t = \sigma_{CR}, \quad (4)$$

where $\sigma_{CR}$ is the experimental $\sigma_y$ of the cold-rolled sample, measured at ~1390 MPa. Thus, the total strengthening contributions of $\Delta\sigma_{gb,CRZ}$, $\Delta\sigma_{d,CRZ}$, and $\Delta\sigma_{t,CRZ}$ are estimated to be about 1172 MPa. According to Eqs. (1)–(4) and the reported parameters [46,47], the total $\sigma_y$ of the present MAs is given by

$$\sigma_y \approx \left(\Delta\sigma_{gb,MPZ} + \Delta\sigma_{d,MPZ}\right) \times f_{MPZ} + \left(\Delta\sigma_{gb,HAZ} + \Delta\sigma_{d,HAZ}\right) \times f_{HAZ} + 218\text{MPa} + 1172\text{MPa} \times f_{CRZ}, \quad (5)$$

On the other hand, $\Delta\sigma_{gb}$ and $\Delta\sigma_d$ in the MPZ and HAZ are determined by the Hall-Petch equation and Taylor's hardening law [48,49], respectively:

$$\Delta\sigma_{gb} = k_y d^{-1/2}, \quad (6)$$

$$\Delta\sigma_d = M\alpha G b \rho^{1/2}, \quad (7)$$

where $k_y \approx 265$ MPa $\mu m^{-1/2}$ is the Hall-Petch slope, $d$ is the average grain size, $M \approx 3.06$ is the average Taylor factor, $G \approx 87$ GPa is the shear modulus [50,51], $\alpha \approx 0.2$ is a constant for the FCC crystals, $b \approx 0.253$ nm is the Burgers vector, and $\rho$ is the total dislocation density. The total dislocations consist of GNDs and statistically stored dislocations (SSDs), with GNDs accounting for ~75 % of the total [52]. Based on EBSD observations, the average grain size and the GND density were roughly measured in the MPZ and HAZ, respectively (Table 1). Meanwhile, the $\Delta\sigma_{gb}$ and $\Delta\sigma_d$ in the MPZ and HAZ were calculated and are listed in Table 1. With increasing the E-beam current, the $\Delta\sigma_{gb}$ and $\Delta\sigma_d$ in the MPZ and HAZ decrease. Thus, the strengthening contributions from the three zones in each sample were calculated using Eqs. (5)–(7), and the calculated $\sigma_y$ of the E1, E2, E3, and E4 samples are approximately 1136 MPa, 881 MPa, 756 MPa, and 506 MPa, respectively. These calculated $\sigma_y$ show a similar trend to the experimental results (Fig. 3(e)), although deviations may arise from errors in the estimated parameters, residual stress induced by EBSM processing or additional HDI strengthening contribution. To accurately determine residual stress, the E4 sample manufactured with the maximum E-beam current was subjected to surface residual stress characterization. For other specimens, a gradual decrease in residual stress is anticipated with reductions in E-beam current. It is demonstrated that the E4 sample exhibits a residual stress of −595 ± 25 MPa. Notably, the surface residual stress in the E4 sample is primarily concentrated in the outermost layer of the MPZ, which measures merely several tens of micrometers in thickness. Given that the total thickness of the tensile sample is 1.85 mm, this stress-affected layer represents less than 2 % of the cross-sectional geometry. Consequently, from a mechanical standpoint, the influence of residual stress on the mechanical properties can be considered negligible.

**Table 1**
The calculated grain size, GND density, $\Delta\sigma_{gb}$, $\Delta\sigma_d$ in the MPZ and HAZ.

| Samples | Zones | Average grain size (μm) | GND denisty ($10^{14}$ $m^{-2}$) | $\Delta\sigma_{gb}$ (MPa) | $\Delta\sigma_d$ (MPa) |
|---|---|---|---|---|---|
| E1 | MPZ | 11.9 ± 5.1 | 0.94 ± 0.48 | 77.0 ± 18.7 | 150.8 ± 86.2 |
| | HAZ | 2.7 ± 1.2 | 1.38 ± 0.67 | 160.3 ± 41.1 | 182.7 ± 120.3 |
| E2 | MPZ | 14.4 ± 5.6 | 1.21 ± 0.41 | 69.8 ± 15.0 | 171.1 ± 73.6 |
| | HAZ | 3.2 ± 1.3 | 1.58 ± 0.54 | 148.1 ± 33.7 | 195.5 ± 96.9 |
| E3 | MPZ | 18.2 ± 6.9 | 0.94 ± 0.43 | 62.2 ± 13.1 | 150.8 ± 77.2 |
| | HAZ | 3.5 ± 1.5 | 1.35 ± 0.61 | 142.7 ± 34.4 | 180.7 ± 109.6 |
| E4 | MPZ | 26.5 ± 8.6 | 0.83 ± 0.37 | 51.5 ± 9.0 | 141.7 ± 66.5 |
| | HAZ | 3.3 ± 1.8 | 1.03 ± 0.39 | 145.4 ± 49.5 | 157.9 ± 70.0 |

### 3.3. Hetero-deformation-induced hardening and plastic deformation

To evaluate the additional HDI strengthening and strain-hardening capability, loading-unloading-reloading (LUR) observations are presented in Fig. 4(a). Based on previous studies [3,12,28,53–55], the calculations of HDI stress ($\sigma_{HDI}$), effective stress ($\sigma_{eff}$), and flow stress ($\sigma_{flow}$) are schematically illustrated in Fig. 4(b). In general, $\sigma_{HDI}$ is an internal stress generated by the splitting of local strain and is associated with long-range interactions of moving dislocations. This stress is strongly influenced by the GND density, implying that higher GND density results in higher HDI stress [3,11,12,28,53–56]. Moreover, $\sigma_{eff}$ represents the local stress required to move dislocations and is related to short-range interactions, which are highly dependent on SSDs. $\sigma_{flow}$ is the sum of both $\sigma_{eff}$ and $\sigma_{HDI}$, reflecting the overall stress required for plastic deformation [3,11,12,28,53–56]. Yang et al. [28] proposed a method to calculate $\sigma_{HDI}$, which is given by

$$\sigma_{HDI} = (\sigma_{rs} + \sigma_{us}) / 2, \quad (8)$$

where $\sigma_{rs}$ and $\sigma_{us}$ represent the reloading and the unloading yield stresses, respectively. As illustrated in Fig. 4(c) and (d), with increasing E-beam current, $\sigma_{HDI}$ and $\sigma_{eff}$ decrease, but $\sigma_{HDI}$ decreases more significantly, indicating the importance of the HDI strengthening during plastic deformation. Samples treated with low E-beam currents exhibit higher $\sigma_{HDI}$ compared to those treated with high E-beam currents, and the slopes of the $\sigma_{HDI}$ increasing with strain are steeper for the latter.

It is well-known that the cold rolling process can introduce a large number of slip bands, deformation twins, and textures in the CRZ [39,40]. The high density of deformation defects around grain boundaries can lead to stress concentration, increasing the $\sigma_{HDI}$ [39,40]. In cold-rolled materials, dislocation density reaches local saturation, making it increasingly difficult to store additional dislocations. This results in a decrease in the strain-hardening rate. After EBSM treatments, the GND density in the MPZ is relatively low (Table 1) and remains almost unchanged with increasing E-beam current. In contrast, the GND density in the HAZ gradually decreases as recrystallization proceeds. Therefore, samples treated with low E-beam currents, which retain a more pronounced deformed structure, cannot achieve continuous dislocation storage. This results in higher back stresses but lower HDI hardening rates. When the deformed structures are almost completely recrystallized, the rate of GND storage in the soft MPZ and the hard HAZ increases, leading to a higher HDI hardening rate. For instance, the steeper slope of the E4 sample is indicative of the enhanced GND accumulation in the MPZ and HAZ.

To examine the strain gradients between soft and hard zones, EBSD measurements were conducted on the E4 sample to assess the grain reference orientation deviation (GROD) and the dislocation accommodation (Fig. 5(a)). Compared to the undeformed E4 sample, the grains in the MPZ and HAZ exhibit different local orientation adjustments, even though the overall grain orientation remains unchanged (Fig. 5(a)). This suggests that dislocation activity and lattice distortion occur within the grains, leading to changes in local orientation. The statistical plots of the average GROD in the MPZ and HAZ (Fig. 5(a)) show that as the strain increases, the GROD in the MPZ rises from about 1.64° to 6.51°, while the average GROD in the HAZ increases from around 1.08°–3.05°. The difference in GROD between the two zones confirms the strain incompatibility between the MPZ and HAZ during plastic deformation. Additionally, when the strain reaches approximately 12 %, a significant strain gradient is observed from the MPZ to the HAZ. Furthermore, the

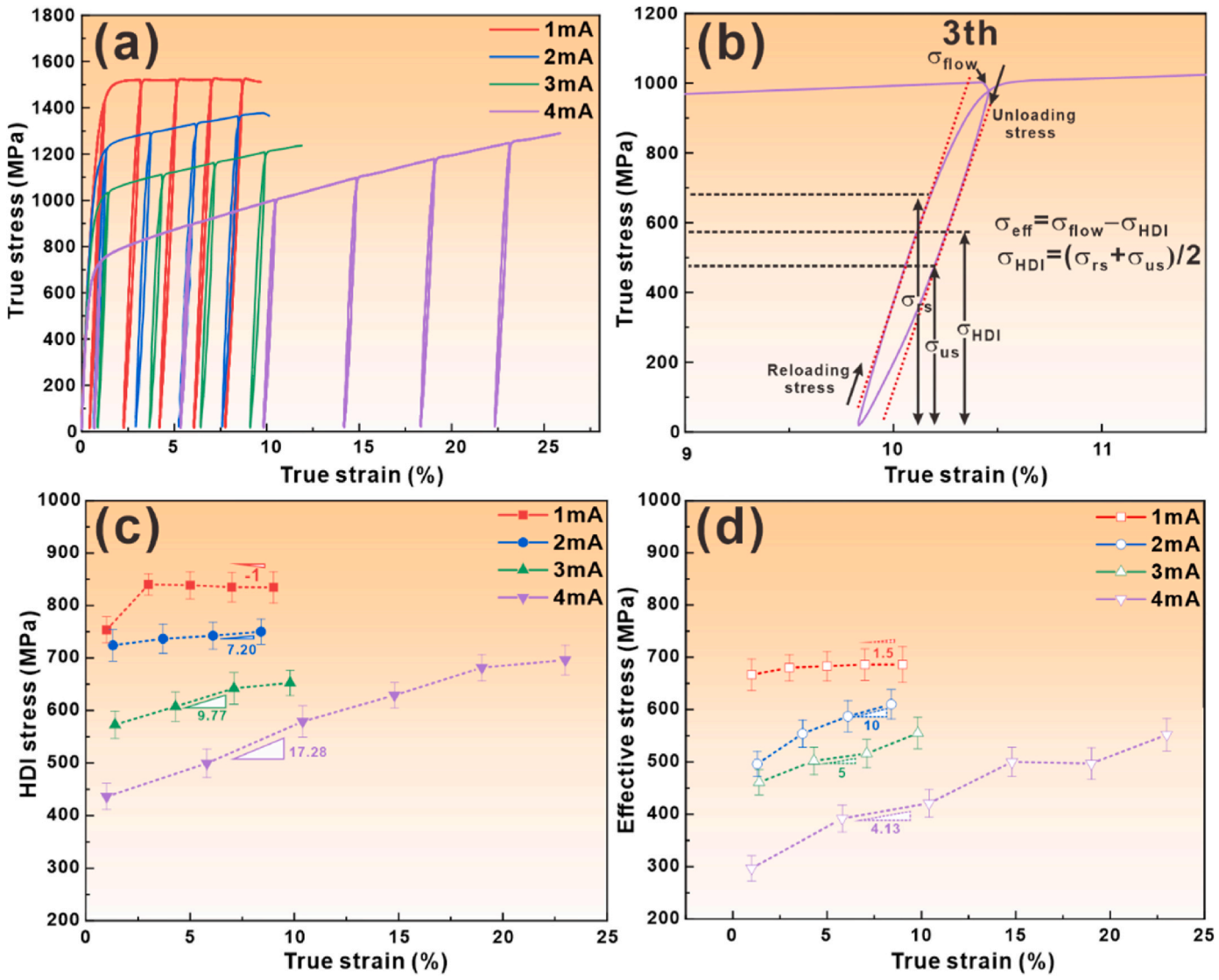



**Fig. 4.** (a) LUR curves after EBSM treatments, (b) calculation of the $\sigma_{HDI}$, $\sigma_{eff}$, and $\sigma_{flow}$, and (c) $\sigma_{HDI}$ and (d) $\sigma_{eff}$ with increasing strain.

evolution of GNDs in the MPZ and HAZ is visible in Fig. 5(b). As the strain increases, the GND density in the MPZ rises from approximately $1.37 \times 10^{14}$ m$^{-2}$ to $3.6 \times 10^{14}$ m$^{-2}$ while the GND density in the HAZ increases from about $2.74 \times 10^{14}$ m$^{-2}$ to $5.41 \times 10^{14}$ m$^{-2}$. The GNDs tend to accumulate around the grain boundaries in the MPZ, whereas they are primarily formed within the grains in the gradient HAZ. Therefore, the plastic deformation is inhomogeneous but continuous, promoting the formation of strain gradients and inducing high back stresses at interfaces or boundaries in the soft MPZ. These back stresses can mitigate the applied shear stress, thereby effectively strengthening the soft MPZ. Conversely, to maintain the stress balance at the interface, an opposing stress, known as forward stress, develops in the hard HAZ. The interplay between back stress and forward stress at the interface leads to an increase in GND density, which generates additional strain hardening and ensures ductility [17,28,29].

## 4. Conclusions

This work utilized the fast EBSM technique to fabricate high-performance CoCrNi MAs with gradient heterogeneous structures and examined their effects on microstructure and mechanical properties. The EBSM-treated alloys exhibit a microstructure consisting of columnar grains in the MPZ and equiaxed grains in the HAZ, in addition to the CRZ. As the E-beam current increases, the grain in the MPZ grows, while the grain size in the HAZ initially increases, and then both zones expand. EBSM treatment significantly enhances the strength-ductility synergy, which is mainly attributed to the formation of gradient-structured GNDs and the contribution of HDI strengthening. The unique properties imparted by the EBSM treatment distinguish it from other surface treatments, providing a promising approach for producing high-performance materials.

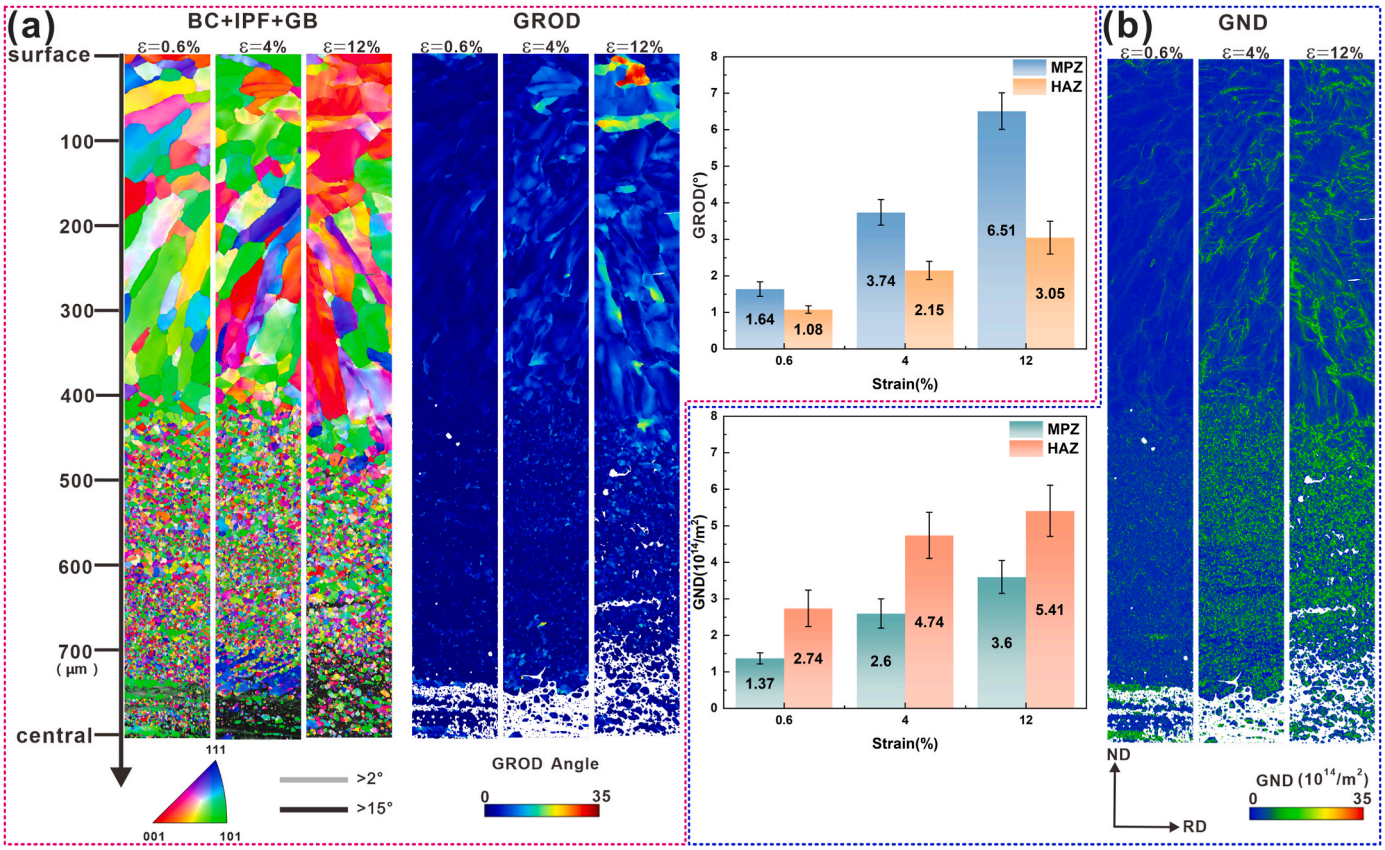



**Fig. 5.** (a) IPF, GROD maps, and statistical average GRODs of the E4 sample under different strains; (b) the distribution and statistical average GNDs in the MPZ and HAZ.

## CRediT authorship contribution statement

**Jilei Zhang:** Writing – original draft, Investigation, Formal analysis, Data curation. **Dekun Si:** Methodology. **Zhenlu Cui:** Data curation. **Wenzhe Zhao:** Methodology. **Xuhao Wang:** Resources. **Zhijie Liang:** Investigation. **Xuran Gong:** Methodology. **Xiaoliang Han:** Formal analysis, Data curation. **Jianhong Gong:** Methodology. **Kaikai Song:** Writing – review & editing, Supervision, Conceptualization. **Zhijie Yan:** Writing – review & editing, Funding acquisition. **Laichang Zhang:** Writing – review & editing, Investigation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

The data that has been used is confidential.

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